



Polar Linear Canonical Wavelet Transform: Theory and Its Application

Hui Zhao¹ · Bingzhao Li²  · Hongmei Zhu³

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Abstract

The polar wavelet transform (PWT) has been proven to be a powerful mathematical tool for signal and image processing in recent years. However, the PWT has some limitations to fully exploit the intrinsic directional features of high-dimensional signals like images. Focusing on this problem, the polar linear canonical wavelet transform (PLCWT) is proposed in this paper. The main goal of this paper is to define PLCWT and investigate its mathematical properties including the inversion formula as well as derive the convolution and correlation theorems of the PLCWT. Finally, a potential application of the PLCWT in medical edge detection is discussed. The ability to detect subtle or weak edges in medical images is the key for accurate early detection and diagnoses. Our numerical experiment demonstrates that the proposed transform PLCWT successfully detects subtle retinal blood vessels of an eye image that were overlooked by both the traditional Canny algorithm and the PWT.

Keywords Linear canonical transform · Polar wavelet transform · Polar linear canonical wavelet transform · Convolution theorem · Edge detection

✉ Bingzhao Li
li_bingzhao@bit.edu.cn

Hui Zhao
zhao_hui2021@163.com

Hongmei Zhu
hmzhu@yorku.ca

¹ Department of Mathematics and Physics, Shijiazhuang Tiedao University, Shijiazhuang 050043, China

² School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China

³ Department of Mathematics and Statistics, York University, Toronto M3J1P3, Canada

1 Introduction

Wavelet transform (WT) is a powerful mathematical analysis tool that has been widely used in many fields, including radar target recognition, signal analysis, image processing and medicine, etc [1, 3, 7, 10, 27, 38, 40, 45]. To effectively analyze and characterize signals, researchers have proposed a series of mathematical transforms, such as fractional Fourier transform (FRFT) [13–15, 37], short-time fractional Fourier transform (STFRFT) [6, 26], linear canonical transform (LCT) [5, 28, 31, 32, 36, 42], fractional wavelet transform (FRWT) [4, 11, 23–25, 39, 44], linear canonical wavelet transform (LCWT) [2, 8, 29, 30], fractional Stockwell transform (FRST) [9, 21, 34], linear canonical Stockwell transform (LCST) [35], etc. However, in medical image processing, the quality of the images is often affected by the limitation of the imaging systems, noises, artifacts and imaging error. The edges of the tissues interested may not be clear and are often overlooked by these transforms due to the subtle differences. The ability to detect subtle or weak edges is the key for accurate early detection and diagnoses. Therefore, it is necessary to do further research to improve the accuracy of edge detection for medical images.

Based on the problem mentioned above, a number of mathematical tools have been proposed, such as the polar wavelet transform (PWT) [12, 22], polar Stockwell transform (PST) [18, 19], and polar offset linear canonical transform [43], etc. These tools have been widely used in image compression, texture analysis, edge detection, and other fields [16, 20, 41]. With the continuous development of information processing technology, high-dimensional signals have become more and more complex in practical applications, and most of them are non-stationary signals. However, the traditional high-dimensional mathematical analysis tools still cannot achieve ideal results in the processing and analysis of these signals. Hence, it is crucial to develop new mathematical tools to effectively address these issues.

In this paper, a new transform method is proposed, namely the polar linear canonical wavelet transform (PLCWT), which combines the advantages of the PWT and LCT. The PLCWT not only has high directional analysis, which improves the efficiency and performance of capturing local information along different angles, but also has the ability to characterize two-dimensional signals on the time-LCT domain plane. The main contributions of this study are summarized as follows.

- (1) To better analyze images in the combined domain of time, direction and the LCT, a novel PLCWT is proposed, and its basic properties are investigated.
- (2) The convolution and correlation theorems are obtained based on the convolution and correlation operators.
- (3) The potential application of the PLCWT in medical image edge detection is discussed. Our numerical experiment demonstrates that the proposed transform PLCWT successfully detects subtle blood vessels of an eye image that were overlooked by both the traditional Canny algorithm and the PWT.

The remainder of this study is structured as follows: Sect. 2 provides some basic preliminaries. In Sect. 3, the theory of the PLCWT is investigated, including its definition, basic properties and inversion formula. In Sect. 4, we present convolution and

correlation theorems for the PLCWT. In Sect. 5, a potential application is presented to demonstrate the importance of the PLCWT. Finally, Sect. 6 concludes the paper.

2 Preliminaries

2.1 Polar Wavelet Transform

In this subsection, we provide some necessary background and notations of the PWT [12, 41].

The polar wavelet is a family of functions constructed from a mother wavelet $\psi \in L^2(\mathbb{R}^2)$ by the combined action of translation, dilation and rotations as [12, 41]

$$\psi_{\mathbf{b},a,\theta}(\mathbf{t}) = \frac{1}{a} \psi \left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a} \right), \quad (1)$$

where $a \in \mathbb{R}^+$, $\mathbf{b} \in \mathbb{R}^2$, $\theta \in [0, 2\pi]$, and R_θ stands for a rotation parameter given by [12, 41]

$$R_\theta(\mathbf{t}) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}, \quad \forall \mathbf{t} = (t_1, t_2) \in \mathbb{R}^2. \quad (2)$$

Let $\psi \in L^2(\mathbb{R}^2)$ be an admissible polar wavelet. The polar wavelet transform (PWT) of any $f \in L^2(\mathbb{R}^2)$ to ψ is denoted by [12, 41]

$$W_f(\mathbf{b}, a, \theta) = \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{t}) \overline{\psi \left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a} \right)} d\mathbf{t}. \quad (3)$$

Or equivalently, it can be rewritten as a conventional convolution

$$\begin{aligned} W_f(\mathbf{b}, a, \theta) &= f(\mathbf{t}) * \left(a^{-1} \overline{\psi(R_{-\theta}(-\mathbf{t}/a))} \right) \\ &= \langle f, \psi_{\mathbf{b},a,\theta} \rangle_{L^2(\mathbb{R}^2)}, \end{aligned} \quad (4)$$

where $*$ and $\langle \cdot, \cdot \rangle$ denote the classical convolution operator and the inner product, respectively.

A polar wavelet is a function $\psi \in L^2(\mathbb{R}^2)$ which satisfies the condition [12, 41]

$$C_\psi = \int_0^\infty \int_0^{2\pi} \left| \mathcal{F}[\psi](a R_{-\theta} \boldsymbol{\xi}) \right|^2 \frac{da d\theta}{a} < \infty, \quad (5)$$

where $\mathcal{F}$ is the polar Fourier transform of ψ . Condition (5) is called the admissibility condition which guarantees the existence of the inversion formula for the continuous polar wavelet transform [12, 41].

2.2 Linear Canonical Transform

The linear canonical transform (LCT) with parameters $M = (A, B; C, D)$ of a signal $f(t) \in L^2(\mathbb{R}^2)$ is defined by [5, 28, 31, 42]

$$F^M(\xi) = \mathcal{L}^M[f(t)](\xi) = \begin{cases} \int_{\mathbb{R}^2} f(t) h_M(t, \xi) dt, & B \neq 0 \\ \sqrt{D} e^{i \frac{CD\xi^2}{2}} \delta(t - d\xi), & B = 0 \end{cases} \quad (6)$$

where the kernel $h_M(t, \xi)$ is

$$h_M(t, \xi) = \frac{1}{2\pi B} e^{j \left[\frac{A}{2B} |t|^2 - \frac{1}{B} t \cdot \xi + \frac{D}{2B} |\xi|^2 \right]}. \quad (7)$$

Here, $\mathcal{L}^M$ denotes the LCT operator, A, B, C , and D are real numbers satisfying $AD - BC = 1$.

The convolution and product theorems of the LCT are introduced in [33]

$$f(t) \Theta_M g(t) = e^{-j \frac{A}{2B} t^2} \left[\left(f(t) e^{j \frac{A}{2B} t^2} \right) * g(t) \right], \quad (8)$$

where Θ_M indicates the linear canonical convolution operator, and $*$ denotes the conventional convolution operator for the Fourier transform (FT). Then, the convolution theorem of the LCT for the signal $f(t)$ and $g(t)$ is given by [33]

$$f(t) \Theta_M g(t) \xleftrightarrow{\mathcal{L}^M} F^M(\xi) G\left(\frac{\xi}{B}\right), \quad (9)$$

where $F^M(\xi)$ and $G(\xi)$ denote the LCT of $f(t)$ and the FT of $f(t)$, respectively.

3 Polar Linear Canonical Wavelet Transform

In this section, a new transform, namely, the polar linear canonical wavelet transform (PLCWT) is proposed, which addresses the limitations of the PWT and LCT. Then, some basic properties are derived. More importantly, it has shown a high degree of directionality compared to previously defined the LCWT.

3.1 Definition

Here, the PLCWT is defined based on the convolution operation in the LCT domain.

Definition 1 (PLCWT) The PLCWT of a signal $f(t) \in L^2(\mathbb{R}^2)$ with parameters $M = (A, B; C, D)$ and polar mother wavelet $\psi \in L^2(\mathbb{R}^2)$ is defined as

$$\begin{aligned} W_f^M(\mathbf{b}, a, \theta) &= f(t) \Theta_M \left(a^{-1} \overline{\psi(R_{-\theta}(-t/a))} \right) \\ &= e^{-j \frac{A}{2B} b^2} \left\langle f(\cdot) e^{-j \frac{A}{2B} (\cdot)^2}, a^{-1} \psi_{\mathbf{b}, a, \theta}(\cdot) \right\rangle_{L^2(\mathbb{R}^2)} \\ &= \int_{\mathbb{R}^2} f(t) \overline{\psi_{\mathbf{b}, a, \theta}^M(t)} dt, \end{aligned} \quad (10)$$

where the transform kernel $\psi_{\mathbf{b}, a, \theta}^M$ satisfies

$$\psi_{\mathbf{b}, a, \theta}^M(t) = e^{-j \frac{A}{2B} (t^2 - b^2)} \psi_{\mathbf{b}, a, \theta}(t), \quad (11)$$

where $\psi_{\mathbf{b}, a, \theta}(t)$ is give by Eq. (1), $a \in \mathbb{R}^+$, $\mathbf{b} \in \mathbb{R}^2$, $\theta \in [0, 2\pi]$. When $M = (0, 1; -1, 0)$, the PLCWT reduced to the PWT.

A polar mother wavelet is a function $\psi \in L^2(\mathbb{R}^2)$ which satisfies the condition

$$C_{\psi, M} = \int_0^\infty \int_0^{2\pi} \left| \mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi) \right|^2 \frac{dad\theta}{a} < \infty. \quad (12)$$

The identity (12) is called the admissibility condition which guarantees the existence of the inversion formula for the PLCWT.

The PLCWT of a signal $f(t)$ in Eq. (10) can be rewritten as

$$W_f^M(\mathbf{b}, a, \theta) = e^{-j \frac{A}{2B} b^2} \int_{\mathbb{R}^2} \left(f(t) e^{j \frac{A}{2B} t^2} \right) \overline{\psi_{\mathbf{b}, a, \theta}(t)} dt. \quad (13)$$

The computation of the PLCWT corresponds to the following steps:

1. A product by a chirp signal, i.e. $f(t) \rightarrow \tilde{f}(t) = f(t) \cdot e^{j \frac{A}{2B} t^2}$;
2. A traditional PWT, i.e. $\tilde{f}(t) \rightarrow W_{\tilde{f}}(\mathbf{b}, a, \theta)$;
3. Another product by a chip signal, i.e.

$$W_{\tilde{f}}(\mathbf{b}, a, \theta) \rightarrow W_f^M(\mathbf{b}, a, \theta) = W_{\tilde{f}}(\mathbf{b}, a, \theta) e^{-j \frac{A}{2B} b^2}.$$

It can be seen from the above steps that the PLCWT can be realized through the discrete algorithm of the PWT. This also means that the computational complexity of the PLCWT depends on the computational complexity of the PWT, and computational complex of the PWT is $O(N)$. Therefore, the computational complexity of the PLCWT is $O(N^2)$. As can be seen from the above, the PLCWT is easy to implement in practice and has low complexity. Furthermore, as shown in Fig. 1, the proposed PLCWT can be decomposed around the PWT embedded in its definition given by Eq. (13).

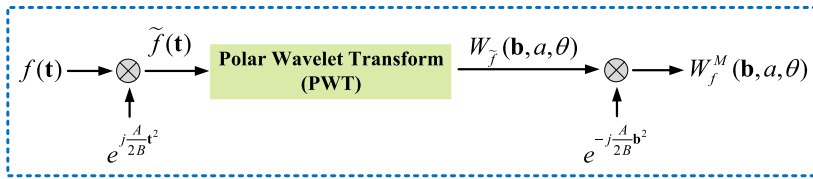


Fig. 1 Structure of computing the proposed PLCWT

3.2 Basic Properties

We then investigate some basic properties of the PLCWT. As a combination of the PWT and LCT, it is straightforward to derive the linearity of the PLCWT, that is,

Property 1 (Linearity) Let $W_h^M(\mathbf{b}, a, \theta)$ be a PLCWT defined by a polar mother wavelet $\psi \in L^2(\mathbb{R}^2)$. If $h(t) = \alpha f(t) + \beta g(t)$, where $f, g \in L^2(\mathbb{R}^2)$, $\alpha, \beta \in \mathbb{C}$. Then

$$W_h^M(\mathbf{b}, a, \theta) = \alpha W_f^M(\mathbf{b}, a, \theta) + \beta W_g^M(\mathbf{b}, a, \theta). \quad (14)$$

Property 2 (Scaling) Let $W_h^M(\mathbf{b}, a, \theta)$ be a PLCWT defined by a polar mother wavelet $\psi \in L^2(\mathbb{R}^2)$. For any $f \in L^2(\mathbb{R}^2)$, if $h(t) = f(ct)$, where $c \neq 0 \in \mathbb{C}$, then

$$W_h^M(\mathbf{b}, a, \theta) = \frac{1}{c} W_f^{M_1}(\mathbf{bc}, ac, \theta), \quad (15)$$

where $M_1 = (A_1, B_1; C_1, D_1)$ satisfies $\frac{A_1}{B_1} = \frac{A}{BC^2}$.

Proof The proof of this property is provided in Appendix A. $\square$

Property 3 (Translation) Let $W_h^M(\mathbf{b}, a, \theta)$ be a PLCWT defined by a polar mother wavelet $\psi \in L^2(\mathbb{R}^2)$. If $h(t) = f(t - \mathbf{c})e^{-j\frac{A}{B}\mathbf{t} \cdot \mathbf{c}}$, where $f \in L^2(\mathbb{R}^2)$. Then

$$W_h^M(\mathbf{b}, a, \theta) = e^{-j\frac{A}{B}\mathbf{b} \cdot \mathbf{c}} W_f^M(\mathbf{b} - \mathbf{c}, a, \theta), \quad \forall \mathbf{c} \in \mathbb{R}^2. \quad (16)$$

Proof The proof of this property is provided in Appendix B. $\square$

Next we prove some lemmas to study other fundamental properties.

Lemma 1 Let $\psi \in L^2(\mathbb{R}^2)$ be a polar mother wavelet. For any $f \in L^2(\mathbb{R}^2)$, then the LCT of $\psi_{\mathbf{b}, a, \theta}^M(\mathbf{t})$ is given by

$$\begin{aligned} \mathcal{L}^M \left\{ \psi_{\mathbf{b}, a, \theta}^M \right\} (\xi) &= a e^{j \left[\frac{A}{2B} \mathbf{b}^2 + \frac{D}{2B} \xi^2 - \frac{1}{B} \mathbf{b} \cdot \xi - \frac{D}{2B} (a R_{-\theta} \xi)^2 \right]} \\ &\quad \times \mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi). \end{aligned} \quad (17)$$

Proof The proof of this property is provided in Appendix C. $\square$

Lemma 2 Let $\psi \in L^2(\mathbb{R}^2)$ be a polar mother wavelet. For any $f \in L^2(\mathbb{R}^2)$, we have

$$W_f^M(\mathbf{b}, a, \theta) = a \int_{\mathbb{R}^2} \frac{e^{j \left[\frac{A}{2B} \mathbf{b}^2 + \frac{D}{2B} \xi^2 - \frac{1}{B} \mathbf{b} \xi - \frac{D}{2B} (a R_{-\theta} \xi)^2 \right]}}{\mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi) \mathcal{L}^M \{ f \} (\xi) d\xi}. \quad (18)$$

Proof The proof of this property is provided in Appendix D. $\square$

Property 4 (Orthogonality Property) Let $W_f^M(\mathbf{b}, a, \theta)$ be a PLCWT defined by a polar mother wavelet $\psi \in L^2(\mathbb{R}^2)$. Then, for any $f, g \in L^2(\mathbb{R}^2)$, we have

$$\int_0^{2\pi} \int_0^{+\infty} \int_{\mathbb{R}^2} W_f^M(\mathbf{b}, a, \theta) \overline{W_g^M(\mathbf{b}, a, \theta)} \frac{d\mathbf{b} da d\theta}{a^3} = C_{\psi, M} \langle f, g \rangle_{L^2(\mathbb{R}^2)}, \quad (19)$$

where $C_{\psi, M}$ is give by Eq. (12).

Proof The proof of this property is provided in Appendix E. $\square$

When $f = g$, Property 4 leads to the following corollary. That is,

Corollary 1 Let $f, g \in L^2(\mathbb{R}^2)$. If $f = g$, then we have

$$\int_0^{2\pi} \int_0^{+\infty} \int_{\mathbb{R}^2} \left| W_f^M(\mathbf{b}, a, \theta) \right|^2 \frac{d\mathbf{b} da d\theta}{a^3} = C_{\psi, M} \langle f, g \rangle_{L^2(\mathbb{R}^2)}. \quad (20)$$

Property 5 (Parseval Formula) Let $\psi \in L^2(\mathbb{R}^2)$ be a polar mother wavelet. For any $f \in L^2(\mathbb{R}^2)$, then

$$\int_{\mathbb{R}^2} \left| \mathcal{L}^M \left\{ W_f^M(\mathbf{b}, a, \theta) \right\} (\xi) \right|^2 d\xi = \int_{\mathbb{R}^2} \left| W_f^M(\mathbf{b}, a, \theta) \right|^2 d\mathbf{b} = \|f\|_{L^2(\mathbb{R}^2)}^2. \quad (21)$$

Proof The proof can be easily obtained. $\square$

Property 6 (Energy Preserving Relation) Let $\psi \in L^2(\mathbb{R}^2)$ be a polar mother wavelet. For any $f \in L^2(\mathbb{R}^2)$, then

$$\int_0^{2\pi} \int_0^{+\infty} \int_{\mathbb{R}^2} \left| W_f^M(\mathbf{b}, a, \theta) \right|^2 \frac{d\mathbf{b} da d\theta}{a^3} = C_{\psi, M} \|f\|_{L^2(\mathbb{R}^2)}^2, \quad (22)$$

where $C_{\psi, M}$ is given by Eq. (12).

Proof The proof can be easily deduced by the Parseval Formula. $\square$

3.3 Inversion Formula

The admissibility condition (12) guarantees the existence of the inversion formula for the PLCWT.

Lemma 3 *The PLCWT of any $f \in L^2(\mathbb{R}^2)$ can be related to the PWT, that is*

$$W_f^M(\mathbf{b}, a, \theta) = e^{-j\frac{A}{2B}\mathbf{b}^2} W_{\tilde{f}}(\mathbf{b}, a, \theta), \quad (23)$$

where $\tilde{f}(\mathbf{t}) = f(\mathbf{t}) \cdot e^{j\frac{A}{2B}\mathbf{t}^2}$.

Proof The proof can be obtained directly from Formula (3). $\square$

Lemma 4 *Let $\psi \in L^2(\mathbb{R}^2)$ be a polar mother wavelet. Then, for every $f \in L^2(\mathbb{R}^2)$, the LCT of the PLCWT is obtained by*

$$\mathcal{L}^M \left\{ W_f^M(\mathbf{b}, a, \theta) \right\}(\xi) = a e^{j\frac{D}{2B}(aR_{-\theta}\xi)^2} \overline{\mathcal{L}^M \left\{ e^{-j\frac{A}{2B}(\cdot)^2} \psi \right\}(aR_{-\theta}\xi)} \mathcal{L}^M \{f\}(\xi). \quad (24)$$

Proof According to the definition of the LCT, we obtain

$$\mathcal{L}^M \left\{ W_f^M(\mathbf{b}, a, \theta) \right\}(\xi) = \frac{1}{2\pi B} \int_{\mathbb{R}^2} W_f^M(\mathbf{b}, a, \theta) e^{j\left(\frac{A}{2B}\mathbf{b}^2 - \frac{1}{B}\mathbf{b} \cdot \xi + \frac{D}{2B}\xi^2\right)} d\mathbf{b}. \quad (25)$$

From Eq. (25), we have

$$\begin{aligned} & \mathcal{L}^M \left\{ W_f^M(\mathbf{b}, a, \theta) \right\}(\xi) \\ &= \frac{1}{2\pi B} \int_{\mathbb{R}^2} \left\{ \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{t}) \psi \left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a} \right) e^{j\frac{A}{2B}(\mathbf{t}^2 - \mathbf{b}^2)} d\mathbf{t} \right\} \\ & \quad \times e^{j\left(\frac{A}{2B}\mathbf{b}^2 - \frac{1}{B}\mathbf{b} \cdot \xi + \frac{D}{2B}\xi^2\right)} d\mathbf{b} \\ &= \frac{1}{2\pi a B} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} f(\mathbf{t}) \psi \left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a} \right) e^{j\left(\frac{A}{2B}\mathbf{t}^2 - \frac{1}{B}\mathbf{b} \cdot \xi + \frac{D}{2B}\xi^2\right)} d\mathbf{b} d\mathbf{t} \quad (26) \\ &= \frac{1}{2\pi B} \int_{\mathbb{R}^2} f(\mathbf{w} + \mathbf{b}) e^{j\left[\frac{A}{2B}(\mathbf{w} + \mathbf{b})^2 - \frac{1}{B}(\mathbf{w} + \mathbf{b}) \cdot \xi + \frac{D}{2B}\xi^2\right]} d(\mathbf{w} + \mathbf{b}) \\ & \quad \times \frac{1}{a} \int_{\mathbb{R}^2} \overline{\psi \left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a} \right)} e^{j\frac{1}{B}\mathbf{w} \cdot \xi} d\mathbf{w} \\ &= a e^{j\frac{D}{2B}(aR_{-\theta}\xi)^2} \overline{\mathcal{L}^M \left\{ e^{-j\frac{A}{2B}(\cdot)^2} \psi \right\}(aR_{-\theta}\xi)} \mathcal{L}^M \{f\}(\xi). \end{aligned}$$

$\square$

Theorem 1 (Inversion Formula) *Let $\psi \in L^2(\mathbb{R}^2)$ be a polar mother wavelet. For any $f \in L^2(\mathbb{R}^2)$, then f can be reconstructed by*

$$f(t) = \frac{1}{C_{\psi, M}} \int_0^{2\pi} \int_0^{+\infty} \int_{\mathbb{R}^2} W_f^M(\mathbf{b}, a, \theta) \psi_{\mathbf{b}, a, \theta}^M(\mathbf{t}) \frac{d\mathbf{b} da d\theta}{a^3}, \quad (27)$$

where $C_{\psi, M}$ is the admissibility condition given by Eq. (12).

Proof According the LCT, we get

$$\begin{aligned} \mathcal{L}^M \left\{ \overline{\psi_{\mathbf{b}, a, \theta}^M} \right\}(\xi) &= \frac{1}{2\pi a B} \int_{\mathbb{R}^2} e^{j \frac{A}{2B} (t^2 - \mathbf{b}^2)} \overline{\psi \left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a} \right)} e^{j \left(\frac{A}{2B} \mathbf{b}^2 - \frac{1}{B} \mathbf{b} \cdot \xi + \frac{D}{2B} \xi^2 \right)} d\mathbf{b} \\ &= \frac{1}{2\pi a B} \int_{\mathbb{R}^2} \overline{\psi \left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a} \right)} e^{j \left[\frac{A}{2B} t^2 - \frac{1}{B} \mathbf{b} \cdot \xi + \frac{D}{2B} \xi^2 \right]} d\mathbf{b}. \end{aligned} \quad (28)$$

Let $\eta = \mathbf{t} - \mathbf{b}$, Eq. (28) can be written as

$$\mathcal{L}^M \left\{ \overline{\psi_{\mathbf{b}, a, \theta}^M} \right\}(\xi) = \frac{1}{2\pi a B} \int_{\mathbb{R}^2} \overline{\psi \left(\frac{R_{-\theta}\eta}{a} \right)} e^{j \left[\frac{A}{2B} t^2 - \frac{1}{B} (t - \eta) \cdot \xi + \frac{D}{2B} \xi^2 \right]} d\eta. \quad (29)$$

Let $z = R_{-\theta}\eta/a$, we have

$$\mathcal{L}^M \left\{ \overline{\psi_{\mathbf{b}, a, \theta}^M} \right\}(\xi) = a e^{j \left[\frac{A}{2B} t^2 + \frac{D}{2B} \xi^2 - \frac{1}{B} t \cdot \xi + \frac{D}{2B} (a R_{-\theta} \xi)^2 \right]} \overline{\mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi)}. \quad (30)$$

Using Lemma 4, we obtain

$$\mathcal{L}^M \left\{ W_f^M(\mathbf{b}, a, \theta) \right\}(\xi) = a e^{j \frac{D}{2B} (a R_{-\theta} \xi)^2} \overline{\mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi)} \mathcal{L}^M \{f\}(\xi). \quad (31)$$

From Eq. (30), Eq. (31), and admissibility condition, we have

$$\begin{aligned}
 & \frac{1}{C_{\psi,M}} \int_0^{2\pi} \int_0^{+\infty} \int_{\mathbb{R}^2} W_f^M(\mathbf{b}, a, \theta) \psi_{\mathbf{b},a,\theta}^M(\mathbf{t}) \frac{d\mathbf{b} da d\theta}{a^3} \\
 &= \frac{1}{C_{\psi,M}} \int_0^{2\pi} \int_0^{+\infty} \left\{ \int_{\mathbb{R}^2} e^{j \left[-\frac{A}{2B} t^2 - \frac{D}{2B} \xi^2 + \frac{1}{B} \mathbf{t} \cdot \xi - \frac{D}{2B} (R_{-\theta} \xi)^2 \right]} \right. \\
 & \quad \times \mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi) a^2 e^{j \frac{D}{2B} (a R_{-\theta} \xi)^2} \\
 & \quad \left. \times \overline{\mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi)} \mathcal{L}^M \{ f \} (\xi) d\xi \right\} \frac{da d\theta}{a^3} \quad (32) \\
 &= \frac{1}{C_{\psi,M}} \int_0^{2\pi} \int_0^{+\infty} \left| \mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi) \right|^2 \frac{da d\theta}{a} \\
 & \quad \times \int_{\mathbb{R}^2} \mathcal{L}^M \{ f \} (\xi) e^{j \left[-\frac{A}{2B} t^2 - \frac{D}{2B} \xi^2 + \frac{1}{B} \mathbf{t} \cdot \xi \right]} d\xi \\
 &= f(\mathbf{t}).
 \end{aligned}$$

□

4 Convolution and Correlation Theorems

Convolution theorem is the most fundamental theory in linear time-invariant (LIT) systems. Hence we also establish the convolution and correlation theorems of the PLCWT.

4.1 Convolution and Correlation Operators

We first give the definitions of convolution and correlation operators.

Definition 2 (Convolution Operators) Let f and g be complex-valued functions defined on $\mathbb{R}^2$,

1. Let $f \in L^2(\mathbb{R}^2)$ and $g \in L^2(\mathbb{R}^2)$. Then the convolution operator $\otimes$ is defined as

$$h(\mathbf{x}) = (f \otimes g)(\mathbf{x}) = \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{x} - \mathbf{t}) d\mathbf{t}. \quad (33)$$

If $g \in L^1(\mathbb{R}^2)$, then the integral h exists almost everywhere, $f \otimes g \in L^2(\mathbb{R}^2)$, and $\|f \otimes g\|_{L^2} = \|f\|_{L^2} \cdot \|g\|_{L^1}$.

2. Let $f \in L^1(\mathbb{R}^2)$, $g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$. It is possible to convolve them in just one variable as follows:

$$h_1(\mathbf{y}, \lambda, \theta) = (f \otimes_1 g)(\mathbf{y}, \lambda, \theta) = \int_{\mathbb{R}^2} f(\mathbf{t}, \lambda, \theta) g(\mathbf{y} - \mathbf{t}, \lambda, \theta) d\mathbf{t}, \quad (34)$$

where $\mathbf{y} \in \mathbb{R}^2$, $a \in \mathbb{R}^+$, and $\theta \in [0, 2\pi]$.

3. Let $\psi_f \in L^1(\mathbb{R}^2)$ and $\psi_g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$ be two admissible polar mother wavelets. Their convolution operation $\otimes^\psi$ is defined as a new function given as the following:

$$\psi_h(\mathbf{t}) = (\psi_f \otimes^\psi \psi_g)(\mathbf{t}) = \int_{\mathbb{R}^2} \psi_f(R_{-\theta}\mathbf{y})\psi_g(R_{-\theta}(R_\theta\mathbf{t} - \mathbf{y})) \, d\mathbf{y}. \quad (35)$$

Definition 3 (Correlation Operators) Let f and g be complex-valued functions defined on $\mathbb{R}^2$. We define the correlation operators.

1. Let $f \in L^2(\mathbb{R}^2)$, $g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$. The correlation operator $\odot$ is defined as

$$h(\mathbf{x}) = (f \odot g)(\mathbf{x}) = \int_{\mathbb{R}^2} f(\mathbf{t})g(\mathbf{x} + \mathbf{t}) \, d\mathbf{t}. \quad (36)$$

When $f = g$, $f \odot g$ is called the autocorrelation of f , and g is called the cross-correlation of f and g .

2. Let $f \in L^1(\mathbb{R}^2)$, $g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$. It is possible to correlate them in just one variable as follows

$$h_1(\mathbf{y}, \lambda, \theta) = (f \odot_1 g)(\mathbf{y}, \lambda, \theta) = \int_{\mathbb{R}^2} f(\mathbf{t}, \lambda, \theta)g(\mathbf{y} + \mathbf{t}, \lambda, \theta) \, d\mathbf{t}, \quad (37)$$

where $\mathbf{y} \in \mathbb{R}^2$, $\lambda \in \mathbb{R}^+$, and $\theta \in [0, 2\pi]$.

3. Let $\psi_f \in L^1(\mathbb{R}^2)$ and $\psi_g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$ be two admissible polar mother wavelets. Their convolution operation $\odot^\psi$ is defined as a new function

$$\psi_h(\mathbf{t}) = (\psi_f \odot^\psi \psi_g)(\mathbf{t}) = \int_{\mathbb{R}^2} \psi_f(R_{-\theta}\mathbf{y})\psi_g(R_{-\theta}(R_\theta\mathbf{t} + \mathbf{y})) \, d\mathbf{y}. \quad (38)$$

Since both the convolution and correlation of two valid wavelets satisfy the necessary admissibility conditions, we can use them to analyze convolution and correlation between two signals. The proofs of convolution and correlation theorems of the PLCWT are given in Sects. 4.2 and 4.3.

4.2 Convolution Theorem

The classical convolution theorem can be described as the time-domain convolution is equivalent to the frequency-domain product, which provides the necessary theoretical basis for the design of the FT domain multiplicative filters. Therefore, do the classical convolution operation and convolution formula still retain the same simple product form under the PLCWT? More specifically, does the generalized convolution operation in one transformation domain still correspond to the product form in another domain? This is the key to PLCWT domain filtering. The PLCWT convolution theorem is given below.

Theorem 2 (Convolution Theorem) *Let $\psi_f \in L^2(\mathbb{R}^2)$ and $\psi_g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$ be two admissible polar mother wavelets. If $h = (f \otimes g)$ and $\psi_h = (\psi_f \otimes \psi_g)$, then the PLCWT of h is given by*

$$W_h^M(\mathbf{b}, a, \theta) = \left(W_f^M \otimes_1 W_g^M \right) (\mathbf{b}, a, \theta), \quad (39)$$

where W_f^M and W_g^M denote the PLCWT of $f \in L^2(\mathbb{R}^2)$ and $g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$, respectively.

Proof The PLCWT of $h(t)$, given by Eq. (10), we get

$$W_h^M(\mathbf{b}, a, \theta) = \frac{1}{a} \int_{\mathbb{R}^2} h(\mathbf{x}) \overline{\psi_h \left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b})}{a} \right)} e^{j \frac{A}{2B}(\mathbf{x}^2 - \mathbf{b}^2)} d\mathbf{x}. \quad (40)$$

According to the definition of the convolution operators in Definition 2, we have

$$h(\mathbf{x}) = (f \otimes g)(\mathbf{x}) = \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{x} - \mathbf{t}) d\mathbf{t}, \quad (41)$$

$$\psi_h \left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b})}{a} \right) = \int_{\mathbb{R}^2} \psi_f(R_{-\theta} \mathbf{y}) \psi_g \left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b} - a\mathbf{y})}{a} \right) d\mathbf{y}. \quad (42)$$

Substituting Eq. (41) and Eq. (42) into Eq. (40), Eq. (40) can be written as

$$\begin{aligned} W_h^M(\mathbf{b}, a, \theta) &= \frac{1}{a} \int_{\mathbb{R}^2} h(\mathbf{x}) \overline{\psi_h \left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b})}{a} \right)} e^{j \frac{A}{2B}(\mathbf{x}^2 - \mathbf{b}^2)} d\mathbf{x} \\ &= \frac{1}{a} e^{j \frac{A}{2B}(\mathbf{x}^2 - \mathbf{b}^2)} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{x} - \mathbf{t}) \overline{\psi_f(R_{-\theta} \mathbf{y})} \\ &\quad \times \overline{\psi_g \left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b} - a\mathbf{y})}{a} \right)} d\mathbf{t} d\mathbf{x} d\mathbf{y}. \end{aligned} \quad (43)$$

Making a change of the variables $\mathbf{m} = \mathbf{x} - \mathbf{t}$ and $\mathbf{n} = \mathbf{b} - a\mathbf{y} - \mathbf{t}$ such that $2\mathbf{n}^2 - 2\mathbf{b} \cdot \mathbf{n} + 2\mathbf{t} \cdot \mathbf{m} = 0$, we obtain

$$\begin{aligned} W_h^M(\mathbf{b}, a, \theta) &= \frac{1}{a^2} e^{j \frac{A}{2B} [t^2 - (\mathbf{b} - \mathbf{n})^2 + \mathbf{m}^2 - \mathbf{n}^2]} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{m}) \\ &\quad \times \overline{\psi_f \left(\frac{R_{-\theta}(\mathbf{t} - (\mathbf{b} - \mathbf{n}))}{a} \right)} \overline{\psi_g \left(\frac{R_{-\theta}(\mathbf{m} - \mathbf{n})}{a} \right)} d\mathbf{t} d\mathbf{n} d\mathbf{m} \\ &= \int_{\mathbb{R}^2} \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{t}) \overline{\psi_f \left(\frac{R_{-\theta}(\mathbf{t} - (\mathbf{b} - \mathbf{n}))}{a} \right)} e^{j \frac{A}{2B} [t^2 - (\mathbf{b} - \mathbf{n})^2]} d\mathbf{t} \\ &\quad \times \frac{1}{a} \int_{\mathbb{R}^2} g(\mathbf{m}) \overline{\psi_g \left(\frac{R_{-\theta}(\mathbf{m} - \mathbf{n})}{a} \right)} d\mathbf{m} d\mathbf{n} \\ &= \int_{\mathbb{R}^2} W_f^M(\mathbf{b} - \mathbf{n}, a, \theta) W_g^M(\mathbf{n}, a, \theta) d\mathbf{n} \end{aligned}$$

$$= \left(W_f^M \circledast_1 W_g^M \right) (\mathbf{b}, a, \theta). \quad (44)$$

□

Corollary 2 *It can be observed that for $M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ in Theorem 2, we have*

$$W_h(\mathbf{b}, a, \theta) = (W_f \circledast_1 W_g)(\mathbf{b}, a, \theta), \quad (45)$$

where W_f and W_g denote the PWT of the functions f and g with wavelets ψ_f and ψ_g respectively.

Corollary 3 *It is interesting to note that by taking $\theta = 0$ in Corollary 2, it degenerates into the classical convolution theorem of the WT [17].*

4.3 Correlation Theorem

Similar to the convolution theory of the PLCWT, we investigate whether some conclusions of the classical correlation theorem can be extended to the PLCWT domain. Using the correlation operator to map two functions from the time domain into the PLCWT domain, we then obtain the correlation theorem of the PLCWT.

Theorem 3 (Correlation theorem) *Let $\psi_f \in L^2(\mathbb{R}^2)$ and $\psi_g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$ be two admissible polar mother wavelets. If $h = (f \odot g)$ and $\psi_h = (\psi_f \odot_\psi \psi_g)$, then the PLCWT of h is given by*

$$W_h^M(\mathbf{b}, a, \theta) = \left(W_f^M \odot_1 W_g^M \right) (-\mathbf{b}, a, \theta), \quad (46)$$

where W_f^M and W_g^M denote the PLCWT of the functions $f \in L^2(\mathbb{R}^2)$ and $g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$, respectively.

Proof Using the definition of the LCT, we have

$$W_h^M(\mathbf{b}, a, \theta) = \frac{1}{a} \int_{\mathbb{R}^2} h(\mathbf{x}) \overline{\psi_h \left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b})}{a} \right)} e^{j \frac{A}{2B} (x^2 - b^2)} d\mathbf{x}. \quad (47)$$

According to the definition of the correlation operators, we obtain

$$h(\mathbf{x}) = (f \odot g)(\mathbf{x}) = \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{x} + \mathbf{t}) d\mathbf{t}, \quad (48)$$

$$\psi_h \left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b})}{a} \right) = \int_{\mathbb{R}^2} \psi_f(R_{-\theta} \mathbf{y}) \psi_g \left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b} + a \mathbf{y})}{a} \right) d\mathbf{y}. \quad (49)$$

Combining Eq. (47), Eq. (48), and Eq. (49), we obtain

$$W_h^M(\mathbf{b}, a, \theta) = \frac{1}{a} e^{j \frac{A}{2B} (x^2 - b^2)} \frac{\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{x} + \mathbf{t})}{\psi_f(R_{-\theta} \mathbf{y}) \psi_g\left(\frac{R_{-\theta}(\mathbf{x} - \mathbf{b} + a \mathbf{y})}{a}\right)} d\mathbf{t} d\mathbf{x} d\mathbf{y}. \quad (50)$$

Let variables $\mathbf{m} = \mathbf{x} + \mathbf{t}$ and $\mathbf{n} = \mathbf{b} - a \mathbf{y} + \mathbf{t}$ satisfy $2\mathbf{n}^2 - 2\mathbf{b} \cdot \mathbf{n} + 2\mathbf{t} \cdot \mathbf{m} = 0$. We have

$$\begin{aligned} W_h^M(\mathbf{b}, a, \theta) &= \frac{1}{a^2} e^{j \frac{A}{2B} [t^2 - (n-b)^2 + m^2 - n^2]} \frac{\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{m})}{\psi_f\left(\frac{R_{-\theta}(\mathbf{t} - (\mathbf{n} - \mathbf{b}))}{a}\right) \psi_g\left(\frac{R_{-\theta}(\mathbf{m} - \mathbf{n})}{a}\right)} d\mathbf{t} d\mathbf{n} d\mathbf{m} \\ &= \int_{\mathbb{R}^2} W_f^M(\mathbf{n} - \mathbf{b}, a, \theta) W_g^M(\mathbf{n}, a, \theta) d\mathbf{n} \\ &= (W_f^M \circledast_1 W_g^M)(-\mathbf{b}, a, \theta). \end{aligned} \quad (51)$$

□

Corollary 4 According to Eq. (46), for $M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, Theorem 3 will be reduced to

$$W_h(\mathbf{b}, a, \theta) = (W_f \odot_1 W_g)(-\mathbf{b}, a, \theta), \quad (52)$$

where W_f and W_g denote the PWT of the functions f and g with wavelets ψ_f and ψ_g , respectively.

Corollary 5 It is interesting to note that by taking $\theta = 0$ in Corollary 4, it degenerates into the classical correlation theorem of the WT [17].

5 Potential Applications of the PLCWT

The PLCWT is an effective processing tool in the fields of medical edge detection, ECG signal detection, and audio signal denoising, etc. Next, we will take medical image edge detection as an example to illustrate the importance of the PLCWT.

5.1 Image Edge Detection

Accurate detection of retinal blood vessels is crucial for identifying early stages of eye diseases, such as macular degeneration, diabetic retinopathy, and glaucoma. In early stages of eye diseases, the abnormality is often subtle. In addition, retinal vessel branches are very thin and images often contain noises and distortion. The abnormality

usually appears as weak edges in the images. High accurate edge detection techniques are of importance for early diagnosis and treatment.

Edge detection techniques are indispensable foundations in eye examination. Therefore, we discuss the edge detection for medical images as a potential application of the PLCWT. More specifically, we aim to detect the blood vessels in a retinal fundus image using the PLCWT. Then we compare its performance with the traditional Canny algorithm and the PWT. Simulation results have validated the superiority of the proposed method in retinal imaging edge detection.

Let $f(x, y)$ is a two-dimensional image, $\mathbf{b} = (b_1, b_2)$. The direct discretization of the PLCWT of $f(x, y)$ is

$$W_f^M(\mathbf{b}, a, \theta_i) = \frac{1}{a} \sum_{x=b_1-\frac{N}{2}}^{b_1+\frac{N}{2}} \sum_{y=b_2-\frac{N}{2}}^{b_2+\frac{N}{2}} f(x, y) \Phi(x, y), \quad (53)$$

in which

$$\Phi(x, y) = \frac{1}{a} e^{-j\frac{A}{2B}[(x^2+y^2)-(b_1^2+b_2^2)]} \psi\left(\frac{R_i(x-b_1, y-b_2)}{a}\right), \quad (54)$$

where N is the rang of convolution, R_i is the rotation of θ_i .

The potential application of the PLCWT in image edge detection mainly has the following steps:

Step 1. Discretization of the PLCWT, as in Eq. (53). The mother wavelet chooses Morlet wavelet; θ_i is used to detect edge patches almost parallel to the chosen direction, where $\theta_i = 0, \frac{\pi}{8}, \frac{\pi}{4}, \frac{3\pi}{8}, \frac{\pi}{2}, \frac{5\pi}{8}, \frac{3\pi}{4},$ and $\frac{7\pi}{8}$.

Step 2. Calculate the PLCWT coefficients, and get the edges of eight directions at different scales. The eight sets of edge segments corresponding to the eight selected rotation angles at different scales can complement each other to form an uninterrupted side.

Step 3. Perform image fusion on the results in Step 2, and obtain an edge detection image by thresholding. In practical applications, it is very important to choose the optimal threshold method. The most suitable threshold method needs to be selected according to the actual situation.

Based on the above discussion, we get the theoretical implementation process of the proposed method on medical edge detection, the schematic diagram is shown in Fig. 2. Next, we verify the superiority of the proposed method through simulation experiments.

5.2 Results of Experiments

Here, we present some simulation results and compare the proposed PLCWT with Canny and PWT [22, 41] in retinal image edge detection. The simulation results clearly demonstrate that the proposed method outperforms the Canny algorithm and PWT in this case. To compute the PLCWT results as shown in Fig. 4d, we used the

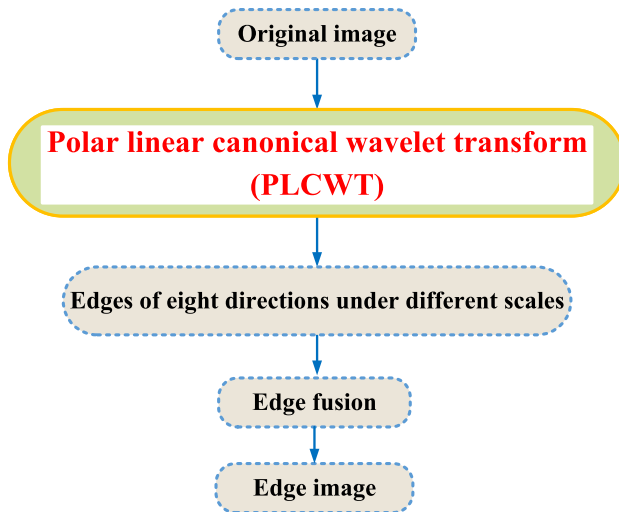


Fig. 2 Schematic diagram of edge detection based on the proposed method

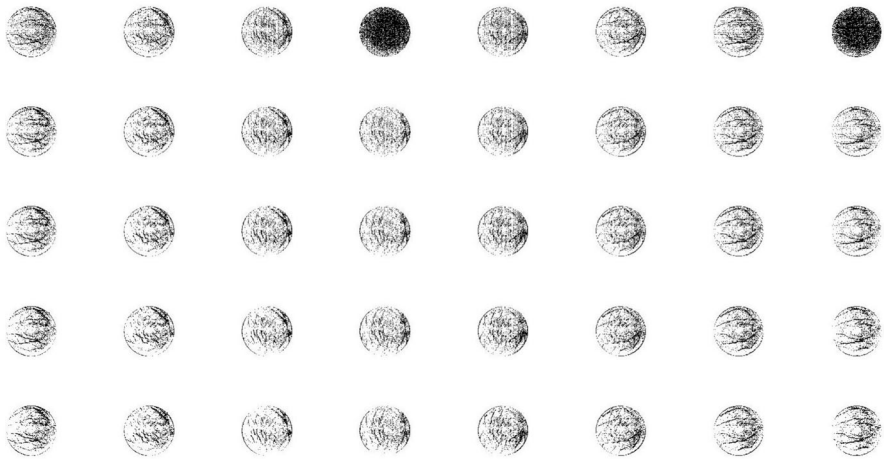


Fig. 3 Edges of eight directions at different scales. From left to right the direction is $0, \frac{\pi}{8}, \frac{\pi}{4}, \frac{3\pi}{8}, \frac{\pi}{2}, \frac{5\pi}{8}, \frac{3\pi}{4},$ and $\frac{7\pi}{8}$

following parameters: $\frac{A}{B} = 0.01$, and the scale factor $a = 1.1$; we selected 5 scale numbers and 8 directions, where $\theta_i = 0, \frac{\pi}{8}, \frac{\pi}{4}, \frac{3\pi}{8}, \frac{\pi}{2}, \frac{5\pi}{8}, \frac{3\pi}{4},$ and $\frac{7\pi}{8}$.

The edges in eight directions at five different scales were extracted from Fig. 4a by PLCWT as shown in Fig. 3. From Fig. 4b–d, we can see that the PLCWT clearly detects more thin blood vessel branches than the Canny algorithm and the PWT. To show the differences more clearly, we enlarged a region of interests (ROI) that contains a big branch of weaker blood vessels (highlighted in a red rectangular box as shown in Fig. 5a, b). This selected branch was missed completely by the Canny algorithm as

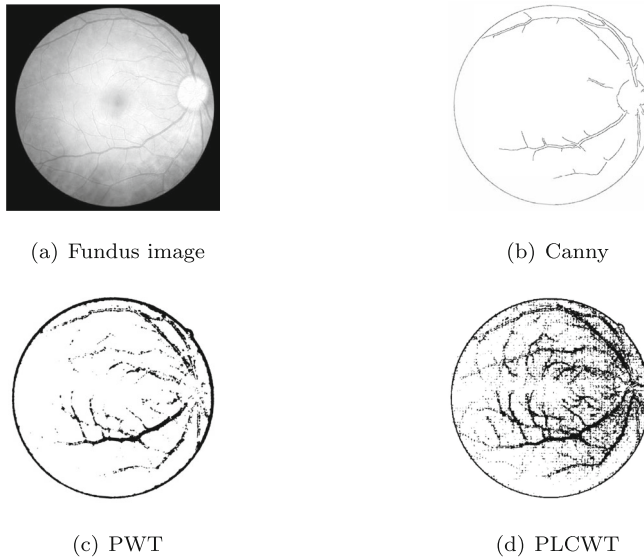


Fig. 4 Identified retinal blood vessels of (a) a fundus image, using (b) Canny, (c) PWT and (d) PLCWT

shown in Fig. 5c, partially by the PWT (as shown in Fig. 5d), but accurately detected by the PLCWT as shown in Fig. 5d).

As shown in Figs. 4 and 5 demonstrate that the PLCWT better preserves the local directional features and edges than the Canny algorithm and the PWT. However, more comprehensive comparison of the PLCWT and the recent transform-based techniques are needed to investigate in the future.

5.3 Discussions

The PLCWT is not only a compelling use case for edge detection in medical imaging, but also has good potential in other areas.

For example, it can be used in analysis of ECG signals for cardiac health monitoring. ECG signals are non-stationary, and detecting abnormalities such as arrhythmia or heart block requires a sensitive and accurate time-frequency analysis tool. Compared to standard methods, the PLCWT can provide higher resolution in the time-frequency domain, allowing better identification of transient and frequency-modulated components in ECG signals, which are often indicative of heart disease, thereby improving the accuracy of cardiac diagnosis.

Furthermore, in the field of audio signal processing, the PLCWT may be advantageous for tasks such as speech enhancement or music analysis. In the context of processing speech signals that are contaminated by background noise, the useful speech components can be more accurately separated from the noisy parts by utilizing the adaptive time-frequency component separation techniques. In the field of music analysis, it can capture the changing frequency modulations and transients in

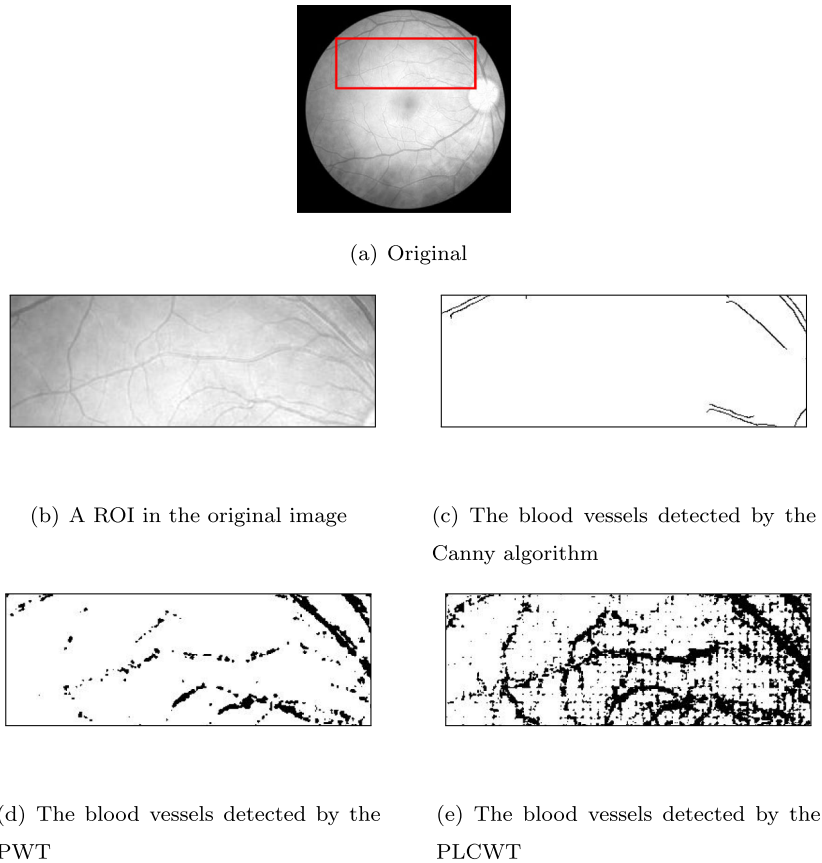


Fig. 5 Performance comparison of the blood vessel detections in a selected ROI of a fundus image of three algorithms: the Canny, the PWT and the PLCWT

musical notes, thereby facilitating tasks such as instrument identification, music transcription and genre classification with a higher degree of accuracy than some of the more commonly used signal processing techniques.

6 Conclusion

In this paper, we proposed a new transform, namely the PLCWT, which integrates the PWT and the LCT. The proposed transform aims to improve the localization of directional features of an image. We first defined the transform and then investigated the basic mathematical properties of the PLCWT including linearity, scaling and inversion formula. We also derived the important convolution and correlation theorems for the PLCWT. Finally, we showed that the PLCWT has potential application in image edge detection. Our numerical experiment showed that the PLCWT detected subtle retinal blood vessels in a fundus image that was overlooked by the Canny and the PWT.

Hence, the PLCWT outperformed the Canny and the PWT, particularly when local directional features are of interests. More comprehensive comparison of the PLCWT and the other commonly used edge detection transforms will be investigated in the future. At present, our work also has some limitations. For example, the PLCWT requires manual adjustment of appropriate parameters, and is sensitive to image noise. In addition, it is computational more intensive than Canny and PWT techniques. In the future, we will look into how to automate the selection of these parameters and overcome its sensitivity to noise by filtering the image first. Faster implementation of the PLCWT is of importance to make it more practical for real-time applications.

Appendix A: Proof of Property 2

Proof From the definition of the PLCWT as in Eq. (10), we have,

$$\begin{aligned}
 W_h^M(\mathbf{b}, a, \theta) &= \left\langle f(c\mathbf{t}), \psi_{\mathbf{b}, a, \theta}^M \right\rangle_{L^2(\mathbb{R}^2)} \\
 &= \frac{1}{a} \int_{\mathbb{R}^2} f(c\mathbf{t}) \overline{\psi\left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a}\right)} e^{j\frac{A}{2B}(t^2 - b^2)} d\mathbf{t} \\
 &= \frac{1}{ac} \int_{\mathbb{R}^2} f(\mathbf{z}) \overline{\psi\left(\frac{R_{-\theta}\left(\frac{\mathbf{z}}{c} - \mathbf{b}\right)}{a}\right)} e^{j\frac{A}{2B}\left(\frac{t^2 - c^2 b^2}{c^2}\right)} d\mathbf{z} \\
 &= \frac{1}{c} \left\{ \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{z}) \overline{\psi\left(\frac{R_{-\theta}(\mathbf{z} - \mathbf{bc})}{ac}\right)} e^{j\frac{A}{2Bc^2}[t^2 - (bc)^2]} d\mathbf{z} \right\} \\
 &= \frac{1}{c} W_f^{M_1}(\mathbf{bc}, ac, \theta),
 \end{aligned} \tag{A.1}$$

where $M_1 = (A_1, B_1; C_1, D_1)$ satisfies $\frac{A_1}{B_1} = \frac{A}{BC^2}$. □

Appendix B: Proof of Property 3

Proof For any $\mathbf{c} \in \mathbb{R}^2$, we get

$$\begin{aligned}
 W_h^M(\mathbf{b}, a, \theta) &= \left\langle f(\mathbf{t} - \mathbf{c}) e^{-j\frac{A}{B}\mathbf{t} \cdot \mathbf{c}}, \psi_{\mathbf{b}, a, \theta}^M \right\rangle_{L^2(\mathbb{R}^2)} \\
 &= \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{t} - \mathbf{c}) e^{-j\frac{A}{B}\mathbf{t} \cdot \mathbf{c}} \overline{\psi\left(\frac{R_{-\theta}(\mathbf{t} - \mathbf{b})}{a}\right)} e^{j\frac{A}{2B}(t^2 - b^2)} d\mathbf{t} \\
 &= \frac{1}{a} e^{-j\frac{A}{B}\mathbf{b} \cdot \mathbf{c}} \int_{\mathbb{R}^2} f(\mathbf{z}) \overline{\psi\left(\frac{R_{-\theta}(\mathbf{z} - (\mathbf{b} - \mathbf{c}))}{a}\right)} e^{j\frac{A}{2B}[z^2 - (b-c)^2]} d\mathbf{z}
 \end{aligned}$$

$$= e^{-j\frac{A}{B}\mathbf{b}\cdot\mathbf{c}} W_f^M(\mathbf{b} - \mathbf{c}, a, \theta). \quad (\text{B.1})$$

□

Appendix C: Proof of Lemma 1

Proof Base on the definition of the LCT as defined in Eq. (10), and let $\mathbf{w} = \mathbf{t} - \mathbf{b}$, we obtain

$$\begin{aligned} \mathcal{L}^M \left\{ \psi_{\mathbf{b},a,\theta}^M \right\} (\boldsymbol{\xi}) &= \frac{1}{2\pi B} \int_{\mathbb{R}^2} \psi_{\mathbf{b},a,\theta}^M(\mathbf{t}) e^{j\left(\frac{A}{2B}\mathbf{t}^2 - \frac{1}{B}\mathbf{t}\cdot\boldsymbol{\xi} + \frac{D}{2B}\boldsymbol{\xi}^2\right)} d\mathbf{t} \\ &= \frac{1}{2\pi a B} \int_{\mathbb{R}^2} \psi\left(\frac{R_{-\theta}\mathbf{w}}{a}\right) e^{j\left[\frac{A}{2B}\mathbf{b}^2 - \frac{1}{B}(\mathbf{w}+\mathbf{b})\cdot\boldsymbol{\xi} + \frac{D}{2B}\boldsymbol{\xi}^2\right]} d\mathbf{w}. \end{aligned} \quad (\text{C.1})$$

Using the change variables $\mathbf{z} = \frac{R_{-\theta}\mathbf{w}}{a}$, we get

$$\begin{aligned} \mathcal{L}^M \left\{ \psi_{\mathbf{b},a,\theta}^M \right\} (\boldsymbol{\xi}) &= \frac{a}{2\pi B} \int_{\mathbb{R}^2} \psi(\mathbf{z}) e^{j\left(\frac{A}{2B}\mathbf{b}^2 - \frac{1}{B}aR_{-\theta}\mathbf{z}\cdot\boldsymbol{\xi} - \frac{1}{B}\mathbf{b}\cdot\boldsymbol{\xi} + \frac{D}{2B}\boldsymbol{\xi}^2\right)} d\mathbf{z} \\ &= \frac{a}{2\pi B} \int_{\mathbb{R}^2} \psi(\mathbf{z}) e^{j\left[-\frac{1}{B}\mathbf{z}\cdot(aR_{-\theta}\boldsymbol{\xi}) + \frac{D}{2B}(aR_{-\theta}\boldsymbol{\xi})^2\right]} d\mathbf{z} \\ &\quad \times e^{j\left[\frac{A}{2B}\mathbf{b}^2 - \frac{1}{B}\mathbf{b}\cdot\boldsymbol{\xi} + \frac{D}{2B}\boldsymbol{\xi}^2 - \frac{D}{2B}(aR_{-\theta}\boldsymbol{\xi})^2\right]} \\ &= a e^{j\left[\frac{A}{2B}\mathbf{b}^2 + \frac{D}{2B}\boldsymbol{\xi}^2 - \frac{1}{B}\mathbf{b}\cdot\boldsymbol{\xi} - \frac{D}{2B}(aR_{-\theta}\boldsymbol{\xi})^2\right]} \\ &\quad \times \mathcal{L}^M \left\{ e^{-j\frac{A}{2B}(\cdot)^2} \psi \right\} (aR_{-\theta}\boldsymbol{\xi}). \end{aligned} \quad (\text{C.2})$$

□

Appendix D: Proof of Lemma 2

Proof From LCT and Lemma 1, we obtain

$$\begin{aligned} W_f^M(\mathbf{b}, a, \theta) &= \left\langle f, \psi_{\mathbf{b},a,\theta}^M \right\rangle_{L^2(\mathbb{R}^2)} \\ &= \left\langle \mathcal{L}^M \{f\}, \mathcal{L}^M \left\{ \psi_{\mathbf{b},a,\theta}^M \right\} \right\rangle_{L^2(\mathbb{R}^2)} \\ &= \int_{\mathbb{R}^2} \mathcal{L}^M \{f\}(\boldsymbol{\xi}) \overline{\mathcal{L}^M \left\{ \psi_{\mathbf{b},a,\theta}^M \right\}(\boldsymbol{\xi})} d\boldsymbol{\xi} \\ &= a \int_{\mathbb{R}^2} e^{j\left[\frac{A}{2B}\mathbf{b}^2 + \frac{D}{2B}\boldsymbol{\xi}^2 - \frac{1}{B}\mathbf{b}\cdot\boldsymbol{\xi} - \frac{D}{2B}(aR_{-\theta}\boldsymbol{\xi})^2\right]} \end{aligned}$$

$$\times \overline{\mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi)} \mathcal{L}^M \{f\}(\xi) d\xi. \quad (\text{D.1})$$

□

Appendix E: Proof of Property 4

Proof By Lemma 2, we get

$$\begin{aligned} & \int_0^{2\pi} \int_0^{+\infty} \int_{\mathbb{R}^2} W_f^M(\mathbf{b}, a, \theta) \overline{W_g^M(\mathbf{b}, a, \theta)} \frac{d\mathbf{b} da d\theta}{a^3} \\ &= \int_0^{2\pi} \int_0^{+\infty} \int_{\mathbb{R}^2} a \int_{\mathbb{R}^2} e^{j \left[-\frac{A}{2B} \mathbf{b}^2 - \frac{D}{2B} \xi^2 + \frac{1}{B} \mathbf{b} \xi + \frac{D}{2B} (a R_{-\theta} \xi)^2 \right]} \mathcal{L}^M \{f\}(\xi) d\xi \\ & \quad \times a \overline{\mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi)} \int_{\mathbb{R}^2} e^{j \left[\frac{A}{2B} \mathbf{b}^2 + \frac{D}{2B} \xi^2 - \frac{1}{B} \mathbf{b} \xi - \frac{D}{2B} (a R_{-\theta} \xi)^2 \right]} \\ & \quad \times \mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi) \overline{\mathcal{L}^M \{g\}(\xi)} d\xi \frac{d\mathbf{b} da d\theta}{a^3} \\ &= \frac{1}{a} \int_0^{2\pi} \int_0^{+\infty} \left| \mathcal{L}^M \left\{ e^{-j \frac{A}{2B} (\cdot)^2} \psi \right\} (a R_{-\theta} \xi) \right|^2 da d\theta \int_{\mathbb{R}^2} \mathcal{L}^M \{f\}(\xi) \overline{\mathcal{L}^M \{g\}(\xi)} d\xi \\ &= C_{\psi, M} \left\langle \mathcal{L}^M \{f\}(\xi), \mathcal{L}^M \{g\}(\xi) \right\rangle_{L^2(\mathbb{R}^2)} \\ &= C_{\psi, M} \langle f, g \rangle_{L^2(\mathbb{R}^2)}. \end{aligned} \quad (\text{E.1})$$

□

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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